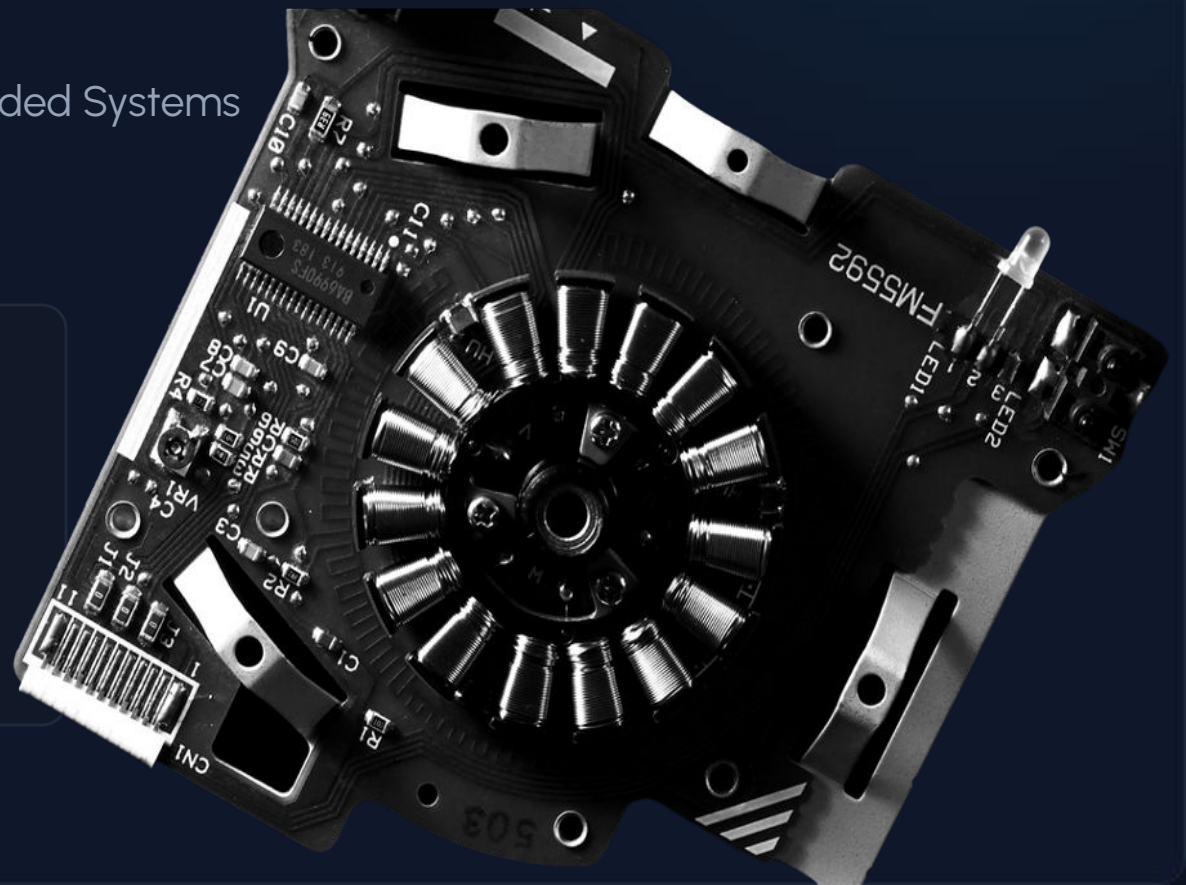
TECHNISCHE HOCHSCHULE DEGGENDORF

Securing Modbus/TCP via SSH Tunneling

A Cryptographic Mitigation Strategy for Legacy Industrial Embedded Systems



Course Context

Selected Topics of Embedded Software Development I

Embedded Security SS2026

Project Team

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Agenda

01	**Introduction & Problem Statement**	Insecurities of Industrial Control Systems (ICS)
02	**Theoretical Background**	Modbus/TCP & Threat Landscape
03	**Phase I Execution**	Plaintext Vulnerability Analysis
04	**Cryptographic Mitigation**	SSH Tunneling Architecture
05	**Phase II Execution**	Secure Tunnel Implementation
06	**Performance Benchmarking**	Feasibility on Embedded Systems
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Introduction to ICS & SCADA

Industrial Control Systems (ICS) and SCADA systems form the digital backbone of modern civilization, regulating power grids, water facilities, and manufacturing floors.

The End of the Air-Gap: Historically, these systems were physically isolated.

Industry 4.0: The Industrial IoT requires real-time data analytics and IT/OT convergence.

The Consequence: This interconnection exposes legacy protocols—designed before modern cyber threats—to the public Internet.

The Modbus/TCP Flaw

The protocol inherently lacks **authentication**, **encryption**, or cryptographic **integrity checks**.

Consequence: High susceptibility to eavesdropping, MITM attacks, and unauthorized data injection.

The Modbus Protocol

Developed in 1979 by Modicon, Modbus is the most widely used industrial protocol. It operates on a strict Master/Slave (Client/Server) architecture.

- **Discrete Inputs & Coils:** Used for reading/writing binary sensors and physical actuators (1-bit).
- **Input & Holding Registers:** Used for reading/writing analog values (16-bits) like temperature or motor speed.
- **Evolution to TCP:** Serial checksums (CRC) were discarded in favor of the TCP/IP stack (Port 502) and an MBAP header.

The Problem & Historical Precedent

Absolute Plaintext

Modbus/TCP transmits all control data and sensor readings in absolute plaintext. It lacks authentication, encryption, or cryptographic integrity checks.

The Stuxnet Context

The 2010 Stuxnet worm demonstrated the catastrophic potential of compromised PLCs. Stuxnet silently modified the control logic of Siemens PLCs governing uranium centrifuges. Lacking integrity checks, the PLCs blindly accepted the malicious logic, resulting in physical destruction.

Threat Landscape for Modbus/TCP

Lack of Auth

No username, password, or handshake. Any device reaching Port 502 can read memory or issue commands immediately.

Eavesdropping

Attackers on the local network (e.g., via ARP spoofing) can read all operational data and deduce production recipes.

Data Injection

Lacking cryptographic signatures (MACs), attackers can alter packets in transit or forge entirely new commands.

Replay Attacks

Without sequence numbers, legitimate commands (e.g., "Start Motor") can be captured and maliciously re-transmitted.

Project Architecture

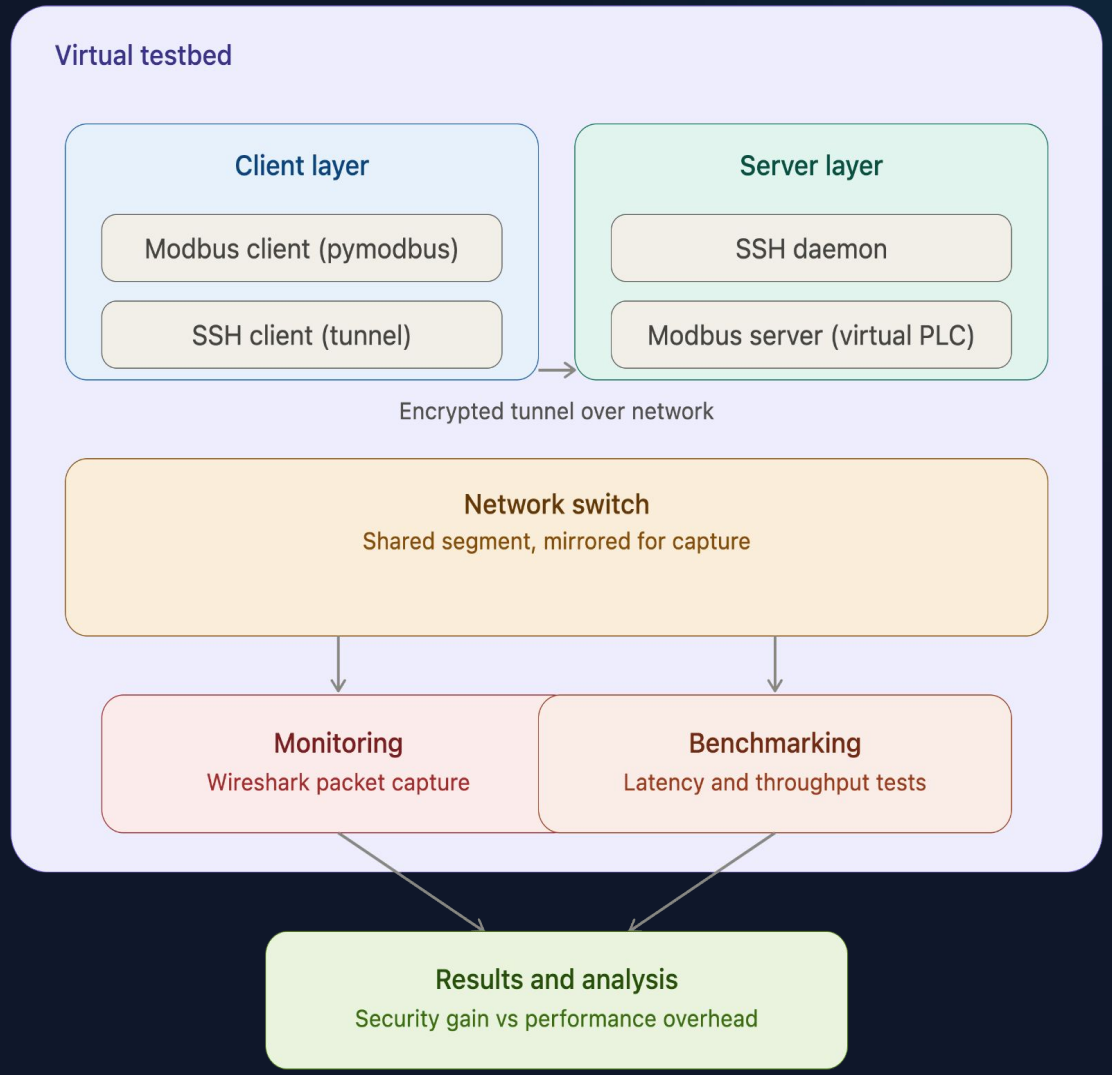
Structured on Clean Architecture, separating industrial logic from the Transport/Security layer.

Virtual PLC (Server)
Uses pymodbus to instantiate an in-memory PLC with simulated registers.

Modbus Client
Acts as the SCADA terminal, continuously reading and writing data.

SSH Orchestrator
Dynamically constructs the cryptographic wrapper using the sshtunnel library.

Makefile Automation
Abstracts Python execution for a unified interface.



Implementation Setup & Workarounds

Port 5020 vs Port 502

UNIX systems prohibit standard users from binding to privileged ports (< 1024). Running the server on Port 502 requires dangerous root access.

Solution: The virtual PLC listens on Port 5020. This circumvents root requirements while preserving exact semantic functionality of Modbus/TCP.

Client-Side Data Generation

A virtual PLC is a "blank brain" without physical sensors. Instead of writing complex background polling threads for the server...

Solution: The Client acts as the simulator, writing simulated Boiler Temps and Motor Speeds to the server, then immediately reading them back. This proves bi-directional encryption later.

Vulnerability Analysis: Base Architecture

Scenario 1: Vulnerable Plaintext

01. System Setup

Direct Connection

A Modbus Client connects directly to a Virtual PLC Server, establishing a standard operational baseline.

02. Clean Architecture

Decoupled Layers

The system layout isolates the core industrial application logic completely from the underlying transport mechanics.

03. Passive Eavesdropping

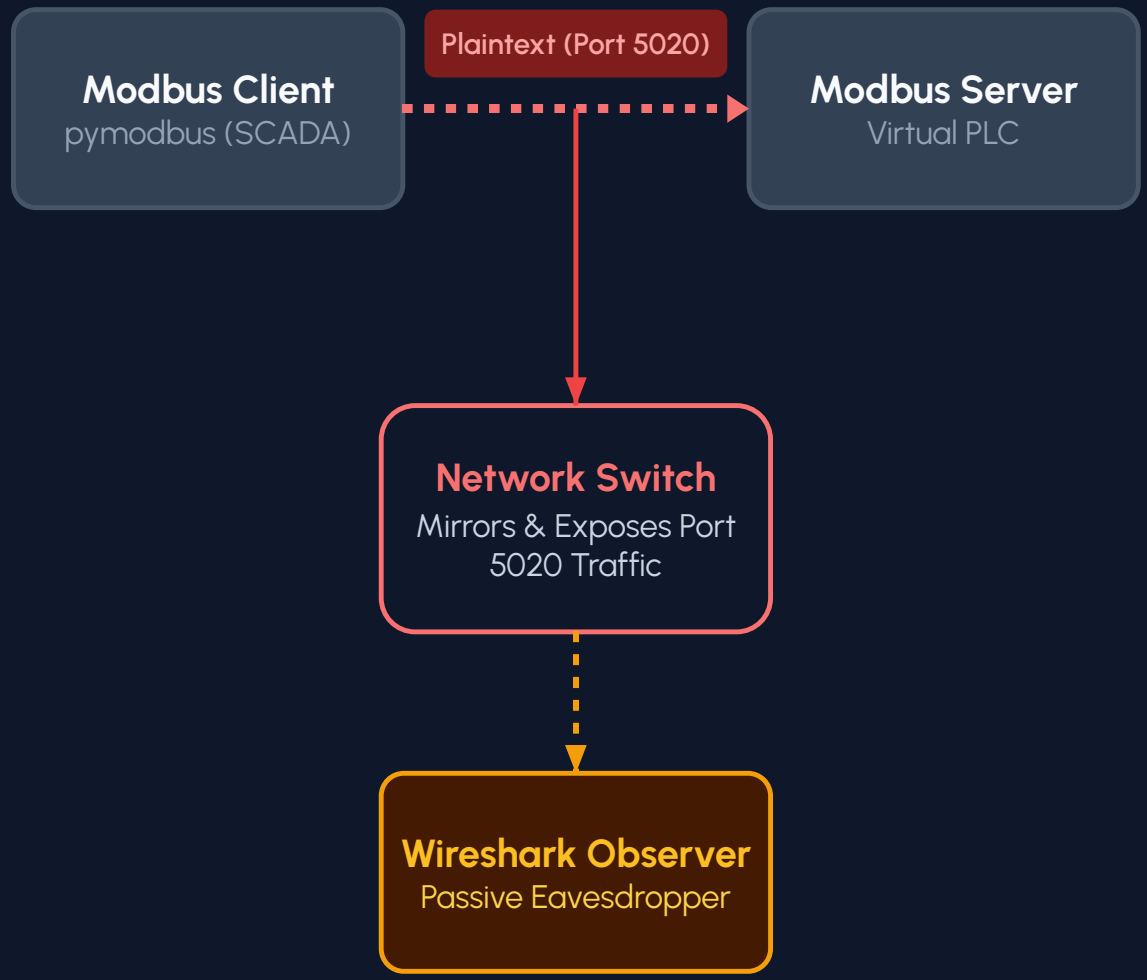
Traffic Mirroring

A network switch transparently duplicates all plaintext traffic flowing on Port 5020 straight to a Wireshark node.

Critical Result

Complete Payload Exposure

Without built-in cryptographic security, industrial control registers and operational parameters remain completely readable by anyone on the network.



Phase I: Plaintext Execution (Base System)

Virtual PLC and Client communicating directly over vulnerable TCP Port 5020.

modbus_server.py

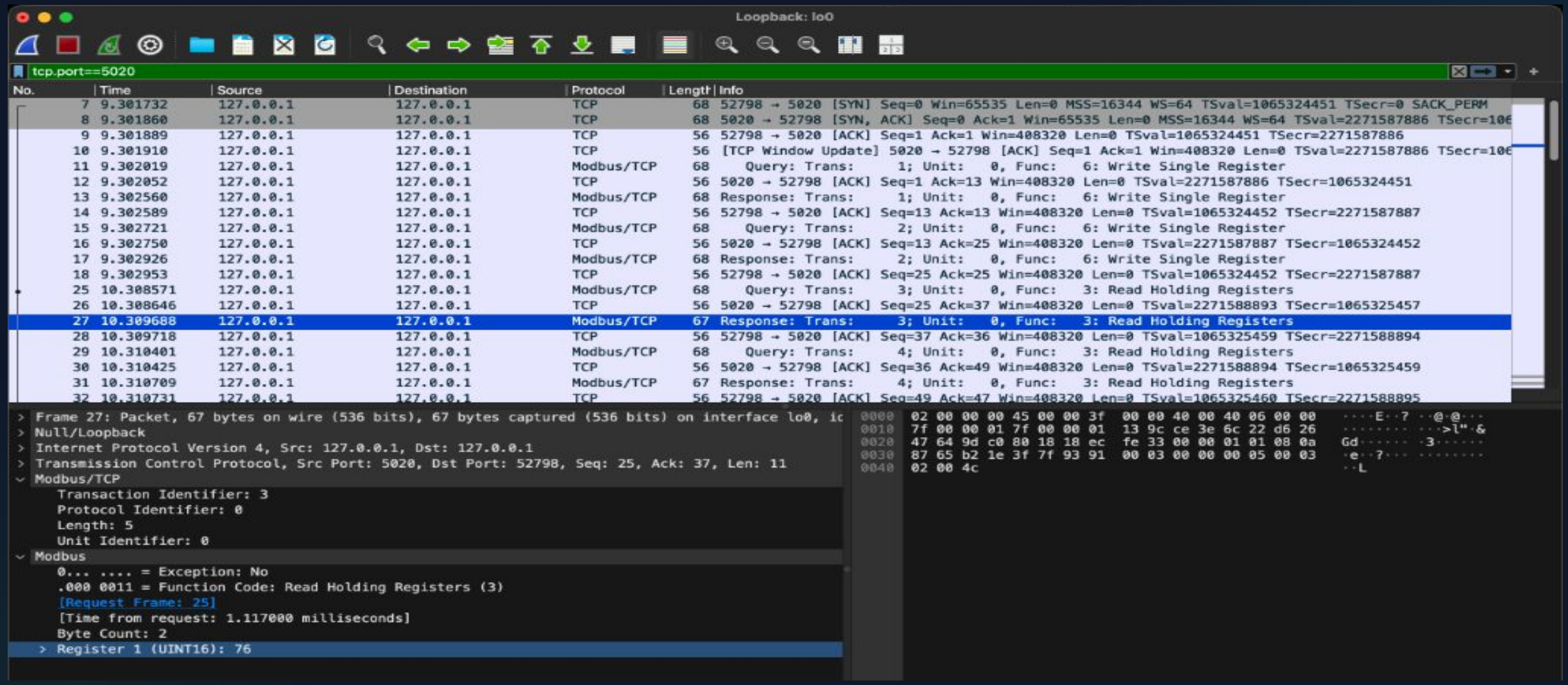
```
Last login: Mon Jun 29 23:48:32 on ttys001
[vedha@MacBookAir ~ % cd "/Users/vedha/Documents/ESD - Modbus:TCP"
[vedha@MacBookAir ESD - Modbus:TCP % make server
python3 src/modbus_server.py
INFO:root:Starting Modbus TCP Server on localhost:5020
INFO:pymodbus.logging:Server listening.
```

modbus_client.py

```
vedha@MacBookAir % make client INFO:root:Connecting to Modbus Server
at 127.0.0.1:5020 INFO:root:Writing Boiler Temp 77°C to register 1
INFO:root:Writing Motor Speed 1489 RPM to register 2 INFO:root:Read
Boiler Temp from register 1: 77°C INFO:root:Read Motor Speed from
register 2: 1489 RPM ──────────────────────────────────────
INFO:root:Writing Boiler Temp 79°C to register 1 ...
```

Phase I: Eavesdropping Vulnerability (Wireshark)

Capturing traffic on the loopback interface reveals industrial data fully exposed in plaintext.



Cryptographic Mitigation: SSH Tunneling

Brownfield Remediation

Replacing legacy PLCs is financially prohibitive. We retrofit security using Secure Shell (SSH) Port Forwarding as a transparent wrapper.

Authentication

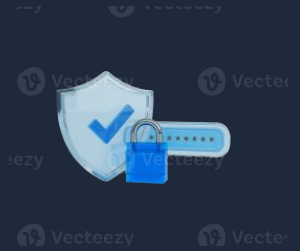
Identity Verification

Public/private key pairs mathematically prove identities before data exchange.

Confidentiality

Data Encryption

AES-256-GCM symmetric encryption scrambles the payload into random noise.



Integrity

Message Signing

Hash-based Message Authentication Code (HMAC) appends a signature. Altered bits cause packets to drop.

Secure System Architecture: SSH Tunneled

Scenario 2: SSH Tunneled Process

01 Tunnel Creation

SSH Client establishes encrypted connection to target Daemon.

02 Local Binding

SSH Client opens local port (localhost:2222).

03 Redirection

Modbus Client points to Port 2222 instead of the distant PLC.

04 Encapsulation

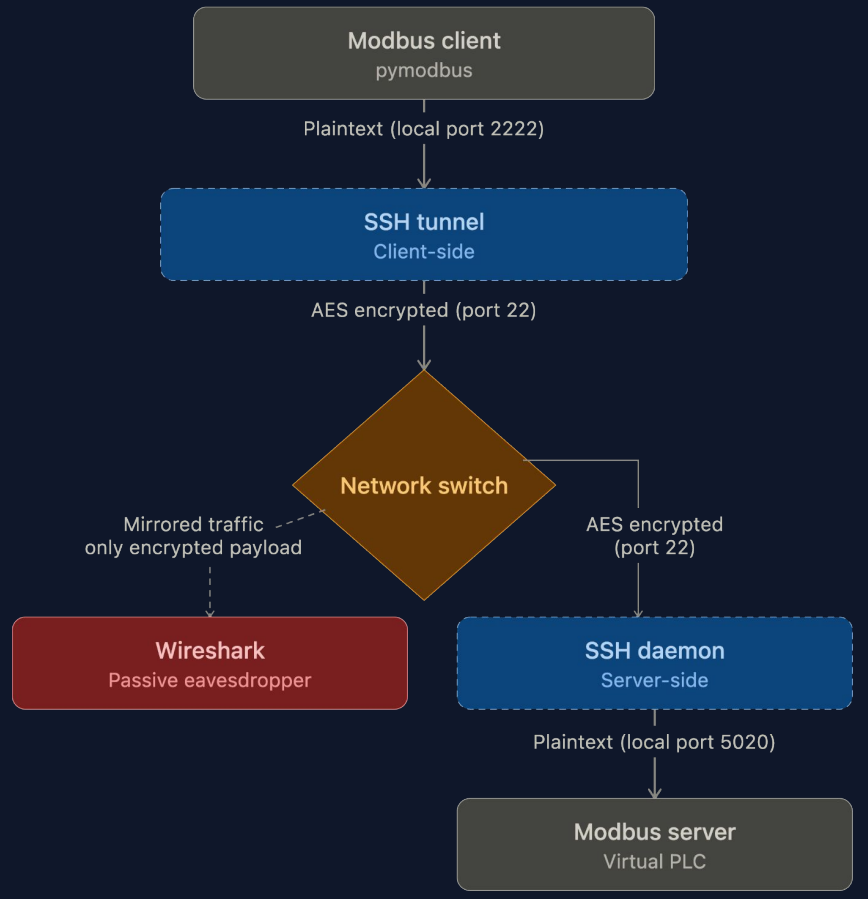
SSH client encrypts traffic via AES and funnels it through Port 22.

05 Decapsulation

Receiving SSH Daemon decrypts and forwards raw bytes to Port 5020.

Result

Application layer remains unaware; encryption occurs transparently at the transport layer.



Result: Modbus payload is protected

The switch and any eavesdropper only see AES ciphertext; register data stays confidential

Phase II: Establishing the Tunnel

The project supports establishing the cryptographic wrapper via native OS tools or a Python library.

Option A: Native macOS SSH

```
vedha@MacBookAir % make tunnel-native Starting Native SSH
Tunnel (Inbuild) on port 2222... ssh -N -L 2222:127.0.0.1:5020
$(whoami)@127.0.0.1 _
```

Option B: Python sshtunnel

```
vedha@MacBookAir % make tunnel-python ═══ Modbus/TCP SSH Tunnel
Setup ═══ Enter your macOS username: vedha Enter your macOS
password:  [+] SUCCESS! SSH Tunnel established. Local Entrance:
Port 2222 Remote Destination: Port 5020
```

Phase II: Secure Client Execution

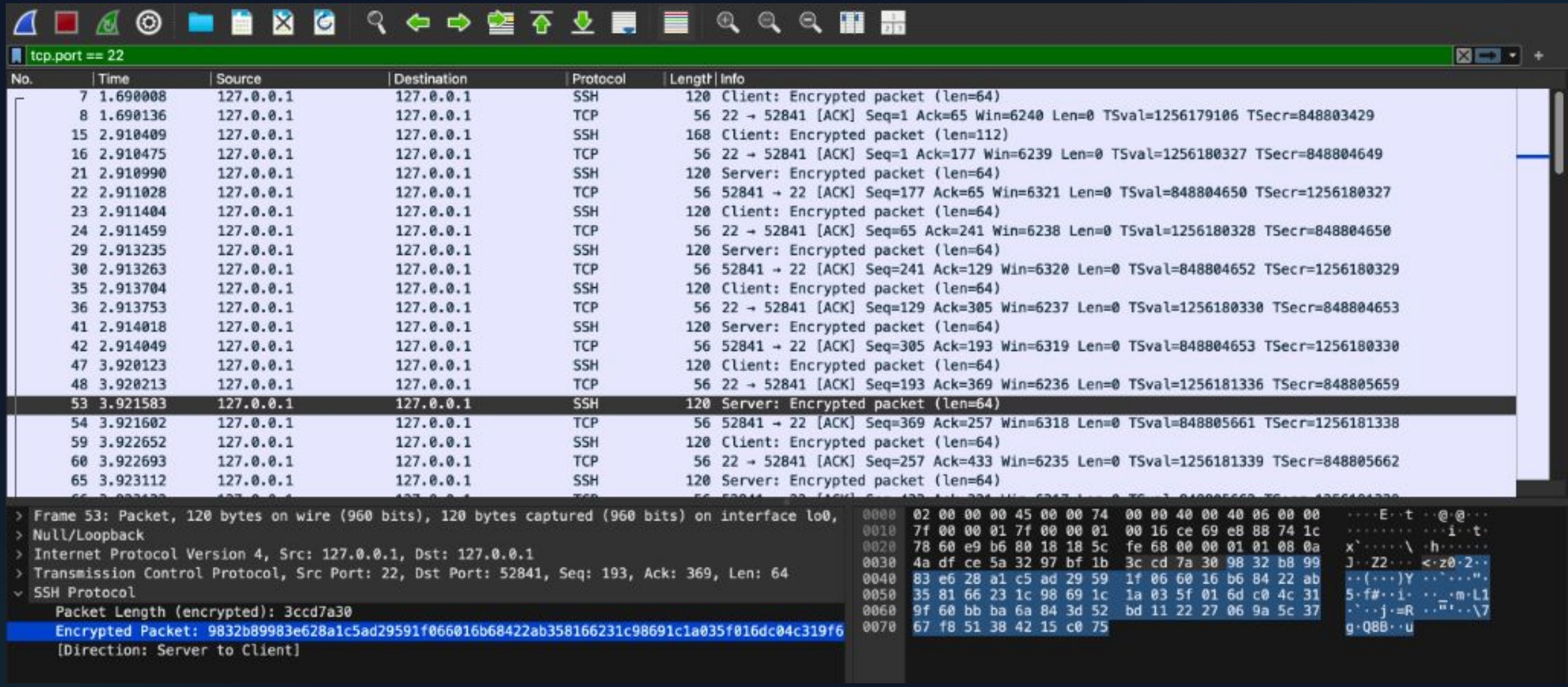
The Modbus client is dynamically re-configured to point its TCP socket to the new local endpoint (Port 2222). Application logic executes flawlessly.

modbus_client.py (Tunnel Mode)

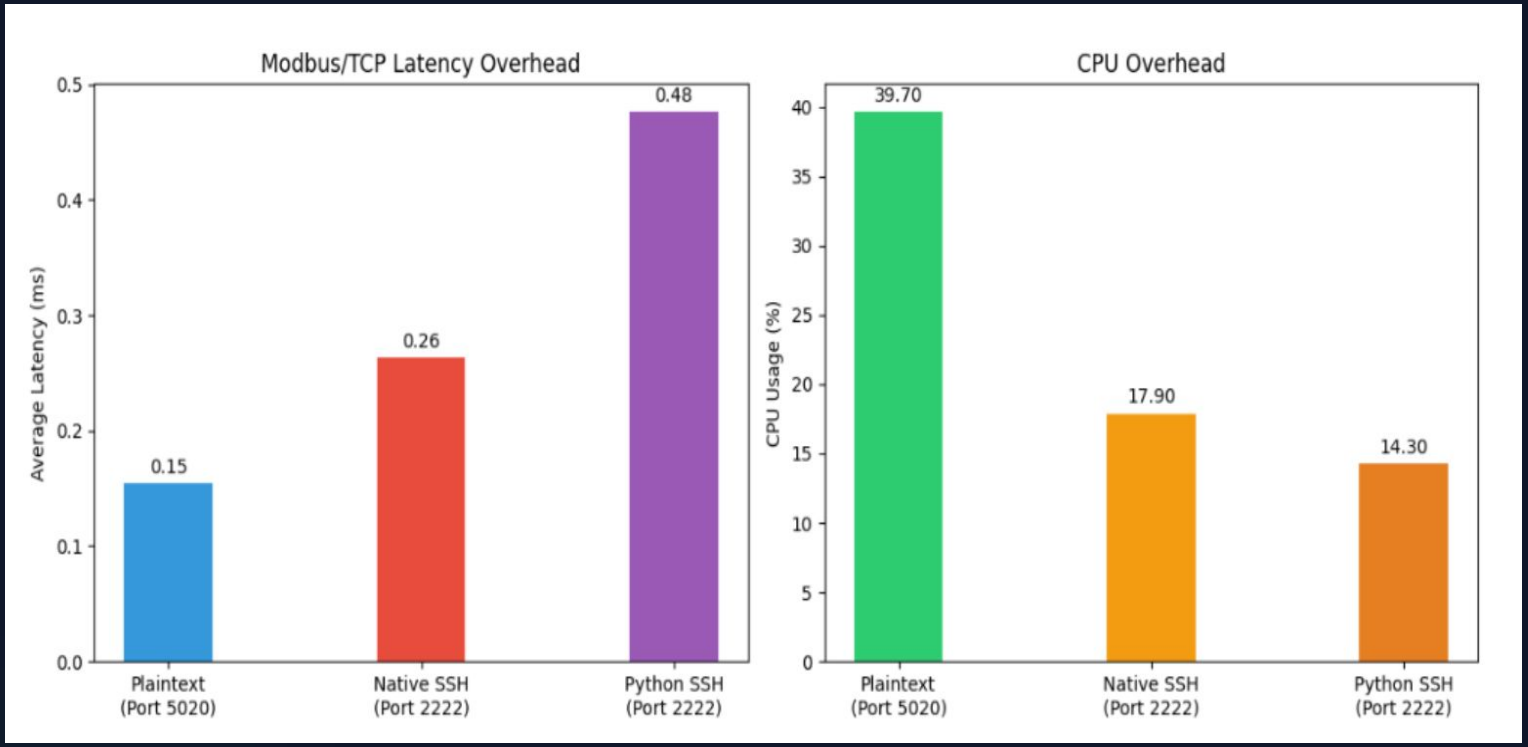
```
vedha@MacBookAir ESD - Modbus:TCP % make client-tunnel python3 src/modbus_client.py --mode tunnel INFO:root:Connecting to
Modbus Server at 127.0.0.1:2222 INFO:root:Writing Boiler Temp 75°C to register 1 INFO:root:Writing Motor Speed 1458 RPM to
register 2 INFO:root:Read Boiler Temp from register 1: 75°C INFO:root:Read Motor Speed from register 2: 1458 RPM
---------------------------------------- INFO:root:Writing Boiler Temp 75°C to register 1 INFO:root:Writing Motor Speed 1477
RPM to register 2 INFO:root:Read Boiler Temp from register 1: 75°C INFO:root:Read Motor Speed from register 2: 1477 RPM
```

Phase II: Blinding the Attacker (Wireshark)

Capturing traffic on the loopback interface reveals industrial data is secure by mathematically random AES sequences. Confidentiality and Integrity are achieved.



Feasibility: Performance Benchmarking



PHASE 1
Plaintext Baseline
(Port 5020)

PHASE 2
Native SSH Tunnel
(Port 2222)

PHASE 3
Python SSH Tunnel
(Port 2222)

```
ESD - Modbus/TCP — -zsh — 142×27
vedha@MacBookAir ESD - Modbus:TCP % cd "/Users/vedha/Documents/ESD - Modbus:TCP"
vedha@MacBookAir ESD - Modbus:TCP % make benchmark
python3 src/modbus_benchmark.py
=== Phase 1: Benchmarking Plaintext Modbus (Port 5020) ===
Executing 1000 consecutive read requests...
  -> Avg Latency: 0.1541 ms
  -> CPU Usage:   39.70%

=== Phase 2: Benchmarking Native SSH Tunnel (Port 2222) ===
Please open a new terminal and run: ssh -N -L 2222:127.0.0.1:5020 $(whoami)@127.0.0.1
Press Enter once the native SSH tunnel is running...
Executing 1000 consecutive read requests...
  -> Avg Latency: 0.2637 ms
  -> CPU Usage:   17.90%

=== Phase 3: Benchmarking Python SSH Tunnel (Port 2222) ===
Please KILL the native SSH tunnel (Ctrl+C), then run: make tunnel-python
Press Enter once the Python SSH tunnel is running...
Executing 1000 consecutive read requests...
  -> Avg Latency: 0.4766 ms
  -> CPU Usage:   14.30%

=== Generating Visualization Graphs ===

[+] Success! Graph saved to 'docs/images/benchmark_results.png'
vedha@MacBookAir ESD - Modbus:TCP %
```

Methodology

Automated Python suite executes 1,000 rapid, consecutive Modbus read requests across three phases to evaluate computational cost on embedded systems.

Future Trends in OT Security

SSH tunneling is an excellent brownfield mitigation, but the future of industrial embedded systems lies in secure-by-design architectures.

- **Protocol Migration:** Shift to OPC UA and MQTT, which integrate TLS encryption natively.
- **Zero Trust Architecture (ZTA):** "Never Trust, Always Verify" - requiring cryptographic identity even on isolated VLANs.
- **Hardware Security (HSM/TPM):** Silicon-level protection for cryptographic private keys.
- **AI-Powered IDS:** Machine learning algorithms on switches to detect anomalous Modbus behaviors.

Conclusion

The Reality: While migrating to natively secure protocols is ideal, legacy infrastructure will remain for decades. Transparent wrappers are an absolute necessity.

KEY VULNERABILITY

Modbus/TCP is critically flawed for interconnected environments.

THE RETROFIT SOLUTION

SSH Tunneling successfully retrofits Confidentiality, Integrity, and Authentication transparently.

FEASIBILITY METRICS

Performance metrics prove it is a highly feasible, cost-effective stopgap for legacy infrastructure.

Questions?

SOURCE CODE REPOSITORY

`github.com/pavaivendhan/ESD-Modbus-SSH-Tunnel`